// NAVEED KHAN BALOCH

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Portfolio · LinkedIn · Google Scholar

PROFESSIONAL SUMMARY

Ph.D. Computer Engineer with **12+ years of experience** in Artificial Intelligence, Computer Vision, Deep Learning, Generative AI, and AI systems engineering. Hands-on experience developing and deploying **object detection, image segmentation, human activity recognition, pose estimation, video analytics, and edge AI** solutions using PyTorch, TensorFlow, OpenCV, MediaPipe, OpenPose, and NVIDIA Jetson platforms.

Experienced in modern **LLMs, RAG, AI Agents, LangChain/LangGraph, multimodal AI, and Generative AI**, with practical experience transforming research concepts into production-oriented AI applications. Strong background in experimentation, model evaluation, AI architecture, API integration, Docker-based deployment, and real-world AI product development.

CORE TECHNICAL EXPERTISE

Computer Vision	Object Detection, Image Segmentation, Video Analytics, Pose Estimation,
CV Libraries	OpenCV, MediaPipe, OpenPose, PyTorch, TensorFlow, Hugging Face
Deep Learning	CNNs, Transformers, GANs, Transfer Learning, Fine-Tuning, Image Enhancement
Generative AI	LLMs, VLMs, RAG, AI Agents, Prompt Engineering, Tool Calling, Multimodal AI
Agentic AI	LangGraph, LangChain, ReAct, Reflection, Multi-Agent Systems, MCP, Tool-Calling
Programming	Python, C++, JavaScript, C#, Java, SQL
Deployment / MLOps	Docker, Docker Compose, REST APIs, CI/CD, Vector Databases, Edge AI
Edge / Hardware	NVIDIA Jetson Xavier, Jetson Nano, Raspberry Pi, Arduino, ESP32
Optimization	ACO, Gray Wolf Optimization, Leader-Follower Algorithms, Nature-Inspired Optimization

PROFESSIONAL EXPERIENCE

AI & IoT Consultant — 2024 – Present
Bluell AB (Sweden) — *Remote*

- **Architected AI-driven automation systems** integrating LLM workflows with IoT infrastructure to enable autonomous, event-driven industrial responses.
- **Engineered real-time Edge AI monitoring solutions** for high-frequency sensor data and intelligent industrial decision-making.
- Consulted on scalable AI architectures for Swedish industrial partners, combining AI orchestration, automation, IoT data pipelines, and intelligent decision systems.

AI Engineer – Top Rated — 2018 – Present
Upwork — *Remote*

- **Architected and developed end-to-end AI applications** spanning Computer Vision, NLP, LLMs, Generative AI, RAG, and Agentic AI for international clients.
- **Developed production-oriented RAG pipelines** using OpenAI, Pinecone, Qdrant, and custom retrieval architectures for semantic search and contextual AI applications.
- **Built Agentic AI systems** using LangGraph and LangChain, implementing tool calling, multi-step reasoning, memory, and workflow orchestration.
- Developed AI-powered voice, interviewing, booking, and business automation systems integrating LLM decision layers with external APIs and automation platforms.

AI & NLP Engineer — 2021 – Present
Skillsoft (USA) — *Remote*

- Architected technical curricula and hands-on learning experiences covering **Generative AI, Machine Learning, NLP, Deep Learning, MLOps, and AI application development** for professional learners.

- Developed practical MLOps modules covering model development, experiment management, containerization, API deployment, and production-oriented AI pipelines.
- Designed technical labs and implementation exercises using modern AI frameworks, Python, PyTorch, and machine learning technologies.

NLP Engineer Jan 2022 – Dec 2022
Synthesis Health (Canada) *Remote*

- Led development of an AI radiology assistant generating clinical "Impressions" from "Findings" using Transformer-based NLP architectures.
- Implemented and evaluated sequence-to-sequence models using modern Transformer architectures and Hugging Face.
- Achieved **12% improvement in ROUGE-L** compared with reported literature benchmarks.
- Implemented human-in-the-loop validation with clinical experts for high-stakes AI output evaluation.

Computer Vision / Generative AI Engineer Mar 2022 – Jan 2023
Augmented Startups *Remote*

- Developed and experimented with **Generative Adversarial Networks (GANs)** for computer vision and image generation applications.
- Implemented **neural style transfer** techniques for automated image transformation and artistic image generation.
- Worked on **image super-resolution and photo enhancement** using deep learning-based approaches.
- Investigated and implemented deep learning techniques for improving visual quality, resolution, and image characteristics.

Co-Principal Investigator – SWARM Robotics Lab Aug 2022 – Present
UET Taxila *Taxila, Pakistan*

- Lead and contribute to **Computer Vision and Edge AI research** for autonomous robotics, drone-based perception, human activity understanding, and intelligent video analytics.
- Developed **object detection systems for drone video streams** and deployed computer vision inference pipelines on NVIDIA Jetson Xavier edge-processing platforms.
- Developed **human activity recognition systems** using MediaPipe and OpenPose for real-time human pose and activity analysis.
- Designed **drone-based head-counting and people detection systems** for aerial video analytics.
- Implemented **image segmentation and object recognition** solutions for robotics and intelligent visual perception applications.
- Developed **vehicle make and model recognition** systems using deep learning and computer vision techniques.
- Developed real-time **exercise simulation and recognition applications** for gym exercises using computer vision and pose estimation.
- Developed **Yoga pose estimation and analysis** applications using human pose detection and computer vision.
- Designed and investigated **nature-inspired optimization algorithms** including Ant Colony Optimization (ACO), Gray Wolf Optimization, and Leader-Follower approaches for swarm robotics coordination and autonomous decision-making.
- Worked across the complete research-to-deployment pipeline, including dataset preparation, model development, experimentation, evaluation, optimization, and deployment on resource-constrained edge hardware.

Assistant Professor 2012 – Present
University of Engineering and Technology (UET) Taxila *Taxila, Pakistan*

- Conduct research and supervise projects in **Artificial Intelligence, Deep Learning, Computer Vision, NLP, Edge AI, and intelligent systems**.
- Supervised **50+ AI-driven student projects** spanning computer vision, healthcare AI, NLP, robotics, and machine learning.
- Published **20+ peer-reviewed research papers** in AI, machine learning, computer engineering, and intelligent systems.

- Designed and deployed AI systems using **PyTorch, TensorFlow, OpenCV, NVIDIA Jetson, and edge computing platforms**.
- Supervise research involving deep learning architectures, computer vision pipelines, optimization algorithms, and AI model evaluation.

SELECTED AI & COMPUTER VISION PROJECTS

Drone-Based Object Detection & Video Analytics

- Developed object detection and head-counting pipelines for aerial drone video and deployed inference on NVIDIA Jetson Xavier for edge processing.

Human Activity & Pose Recognition

- Developed human activity recognition, Yoga pose estimation, and exercise recognition systems using MediaPipe, OpenPose, and deep learning-based computer vision.

Vehicle Make & Model Recognition

- Developed deep learning-based visual recognition systems for identifying vehicle make and model from images/video.

Image Generation, Style Transfer & Enhancement

- Implemented GAN-based applications, neural style transfer, image enhancement, and super-resolution techniques for computer vision applications.

RecrubotX – Agentic AI Hiring SaaS

- Built an end-to-end Agentic AI platform using LangGraph for autonomous CV parsing, candidate evaluation, and voice-based interviews.

Bokafy – Multi-Channel AI Booking Agent

- Engineered a RAG-based booking system using Node.js, Supabase, and LLM decision layers for real-time availability and customer interaction.

Doctor BotX – Voice AI Clinical Assistant

- Developed a voice-integrated clinical assistant using Deepgram/Whisper and Qdrant-based medical knowledge retrieval.

EDUCATION

PhD in Computer Engineering 2020
University of Engineering and Technology (UET), Taxila

MSc in Computer Engineering 2012
University of Engineering and Technology (UET), Taxila

BSc in Computer Engineering 2007
University of Engineering and Technology (UET), Taxila

RESEARCH & PROFESSIONAL SERVICE

- **Co-Principal Investigator:** SWARM Robotics Lab, UET Taxila, leading and contributing to AI, Computer Vision, Edge AI, and swarm robotics research.
- **Research:** 20+ peer-reviewed publications covering AI, machine learning, computer engineering, intelligent systems, and related areas.
- **Reviewer:** IEEE Access, Artificial Intelligence Review, and other indexed Q1 journals.
- **Professional Member:** Pakistan Engineering Council (PEC).

ADDITIONAL TECHNICAL STACK

AI Models	CNN, RNN, LSTM, GAN, Transformers, Vision Models, Sequence-to-Sequence Models
LLM Ecosystem	OpenAI, Claude, Gemini, Llama, Hugging Face, Whisper
Vector / Data	Pinecone, Qdrant, Supabase, PostgreSQL, SQL
Cloud / Deployment	Docker, Docker Compose, AWS SageMaker, REST APIs, CI/CD
Automation	n8n, Make.com, Zapier, API Integrations, Webhooks
Hardware	NVIDIA Jetson Xavier, Jetson Nano, Raspberry Pi, Arduino, ESP32